

Mass Sensitivity Study

Taking a look at contributions

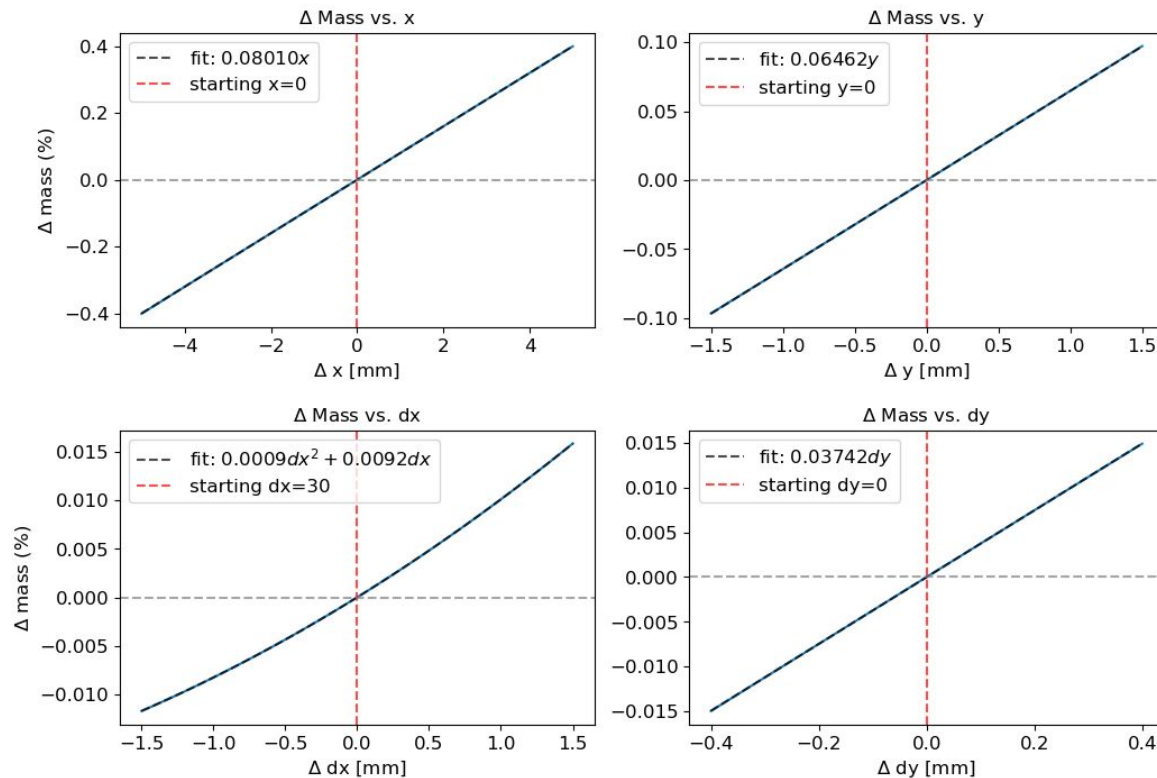


Need to determine how changes in the detector values and change in vertex values affect the mass values.

Very straightforward, vary values within 5% of “nominal” range while keeping other values static and see what happens.

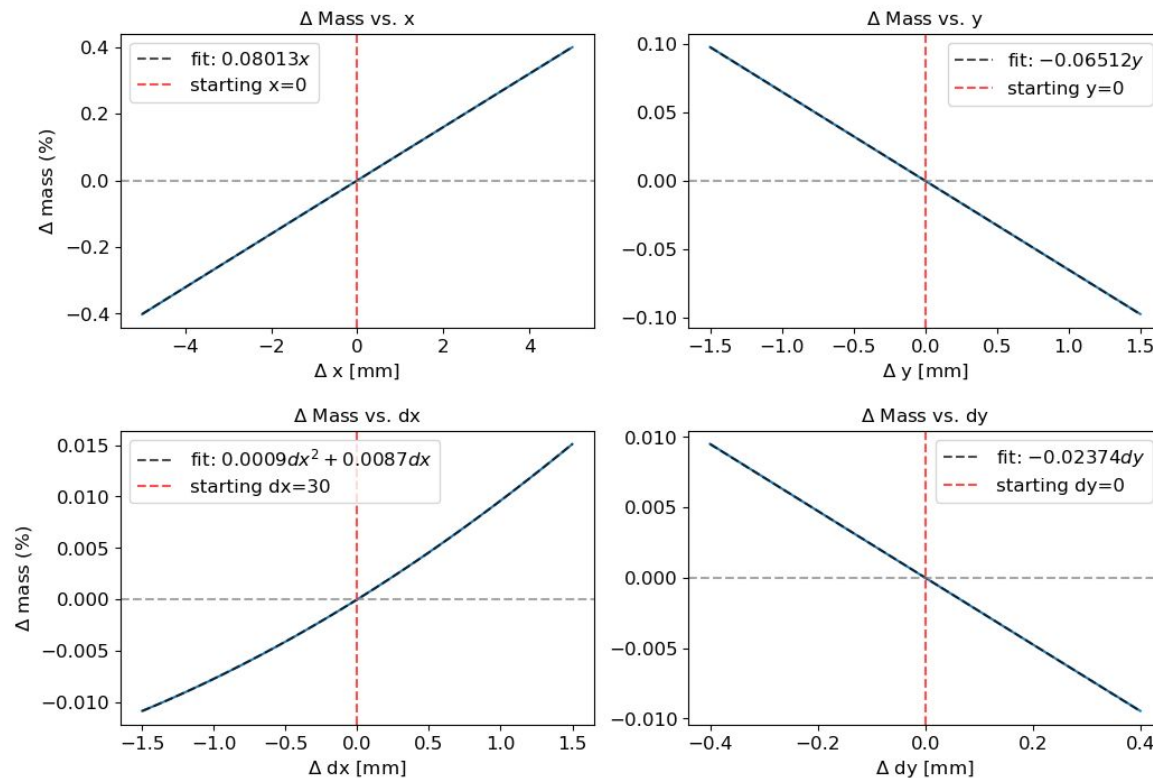
Detector (E) -> mass

Mass sensitivity Electron Arm

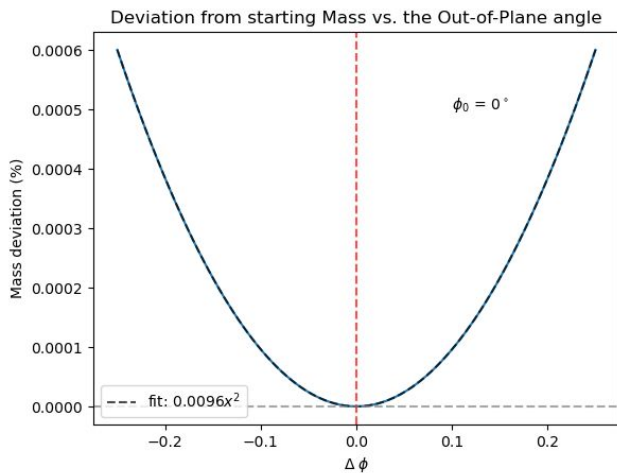


Detector (P) -> mass

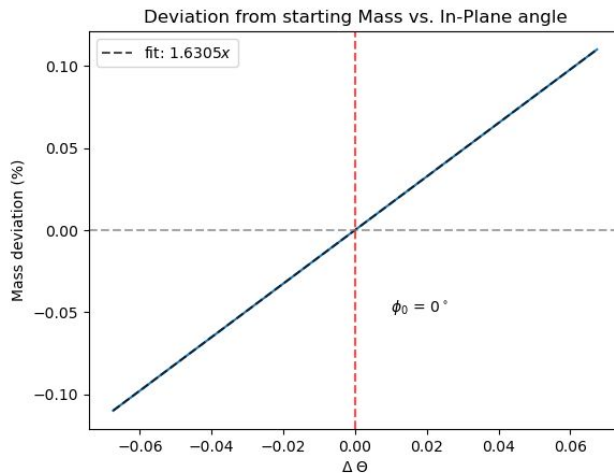
Mass sensitivity Positron Arm



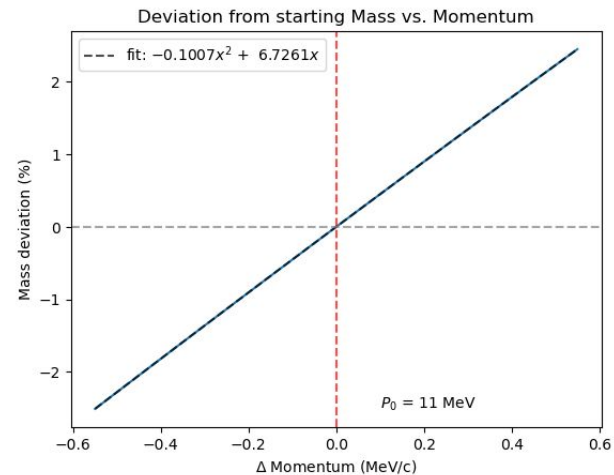
Vertex (E)->mass



$dm/d\phi \sim 0.009\% / \text{degree}$

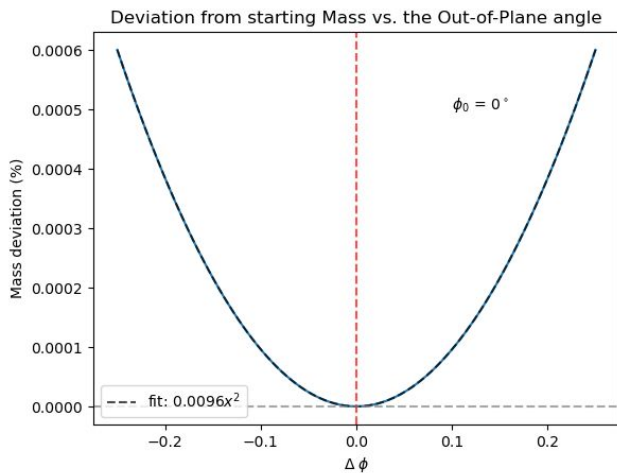


$dm/d\theta \sim 1.63\% / \text{degree}$

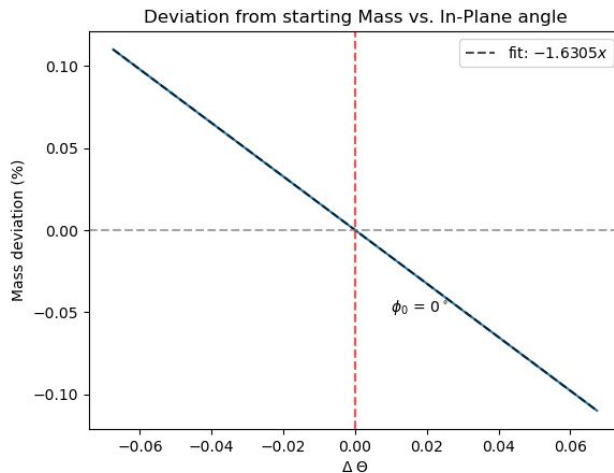


$dm/dp \sim 6.73\% / [\text{MeV/c}]$

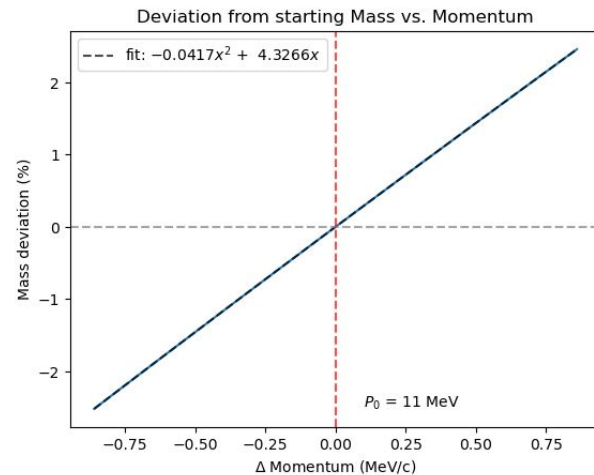
Vertex (P)->mass



$dm/d\phi \sim 0.009\% / \text{degree}$



$dm/d\theta \sim 1.63\% / \text{degree}$



$dm/dp \sim 4.33\% / [\text{MeV/c}]$

Calculated uncertainty approximation



We can use the slope values to approximate the max uncertainty in mass given we have the fit uncertainties, which we do. By summing in quadrature we can derive the approximate resolution we expect $\sim 0.86\%$ with msc + tgt